Triage-PiEvo: Source-Aware Anomaly Diagnosis before Principle Revision

Anonymous authors
Paper under double-blind review

Abstract

Principle-evolving scientific agents can move beyond incomplete initial hypotheses by revising the principles that guide discovery. Yet a large residual is ambiguous: it may reveal a new mechanism, or arise from evaluator corruption, surrogate mismatch, or a local effect that fails to generalize. We introduce Triage , a source-aware evidence-admission layer that separates anomaly detection from authorization to revise persistent scientific state. Triage maintains distinct primary and verification histories, updates channel reliability from reconciliation outcomes, and keeps a proposed principle provisional until it receives independent held-out support; contradictory held-out evidence retracts the principle and its descendants. We instantiate this protocol in Triage-PiEvo , pairing Triage with an outcome-blind Active Flow Expansion portfolio for exploitation, frontier expansion, and mechanism-discriminating probes. Across four matched discovery tasks and three backbone LLMs, Triage-PiEvo achieves the highest average solution quality and area under the optimization curve for each backbone, reaching 79.63–83.37 average solution quality and improving over the strongest baseline for each backbone by 7.5 to 11.8 points. A component study separates the two contributions: search expansion recovers clean-task quality, whereas the admission gate is decisive under unreliable feedback. In controlled corruption, hidden-source, reversal, and proxy-exploitation studies, the gate reduces corrupted-evidence admission from 1.0 to 0.0, rejects the tested non-generalizing artifacts without missing the tested true mechanism, and enables retraction after reversal. The results show that an observation can be useful for directing search before it is reliable enough to justify persistent principle revision.

1 Introduction

Autonomous scientific agents now automate scientific research within a single closed loop (Lu et al., 2024; Yamada et al., 2025; Schmidgall et al., 2025; Gottweis et al., 2025). On the premise that true knowledge often lies outside the initial hypothesis space, principle-evolution agents (Pu et al., 2026; 2025) make every anomaly a decision in this loop: steer hypothesis search, or rewrite the persisted theory and explanation.

While surprise-driven revision (Pu et al., 2026) lets agents escape their initial hypothesis space, it treats the observation's reliability as given. Consequently, the decision rests on a signal that cannot discriminate—the residual magnitude of surprisal may stem from corrupted measurements or surrogate misspecification; thresholding it therefore trades false revisions against missed theory blindly. Nor do existing methods cover this gap: verification loops critique

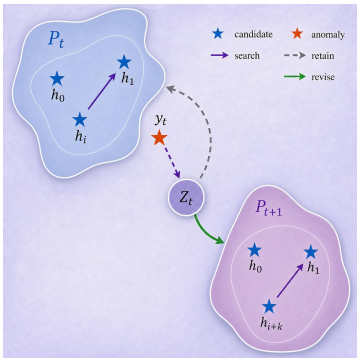


Figure 1: Source-aware triage (schematic). Latent source Z_t mediates the verified retain-or-revise decision.

transient responses rather than persisted theory (Shinn et al., 2023; Madaan et al., 2023; Gou et al., 2023; Dhuliawala et al., 2024), and self-correction without external evidence is unreliable (Huang et al., 2024; Kamoi et al., 2024). Accordingly, community evaluations find that auto-research agents produce extensive artifacts yet seldom revise their internal theory under contradictory evidence (Ríos-García et al., 2026; Kirgis et al., 2026). This rigidity leads to significant inefficiencies: anomalies are read as evidence of science, corrupted feedback enters trusted scientific state.

These limitations reduce to three challenges: (a) one anomaly score has source ambiguity that cannot tell a new mechanism from corrupted or misspecified evaluations; (b) the role conflation of one history serves both search and principle acceptance, so unreliable observations can rewrite trusted principles; and (c) the revision irreversibility that accepted principles cannot be retracted under contradictory evidence.

To bridge this gap, we propose Triage , an evidence-admission layer that decouples anomaly detection from principle revision. Triage infers over a latent source variable for each anomaly and routes evidence along two paths with distinct standards: search outcomes accumulate in a primary history that steers candidate selection, while verification records—same-channel repeats, cross-channel checks, held-out probes—are logged separately and update per-channel reliability estimates. A proposed principle is accepted only after independent held-out support; contradictory held-out evidence retracts it and its descendants. An observation can thus remain search-active while its scientific interpretation stays provisional. We instantiate this protocol in Triage-PiEvo , which pairs Triage with an outcome-blind Active Flow Expansion (AFE) portfolio: the portfolio allocates the evaluation budget across exploitation, frontier expansion, and mechanism-discriminating probes, while the gate alone decides which evidence may alter principle state.

We evaluate Triage-PiEvo on four discovery tasks under three backbone LLMs. Empirical results demonstrate that Triage-PiEvo : (a) attains the highest performance, raising average solution quality from PiEvo 's 62.87–70.35% to 79.63–83.37%; (b) admits no corrupted observations where gate-off variants admit every one (admission rate 0.000 vs. 1.000) under transient and persistent evaluator corruption; (c) cuts false expansions on noise, surrogate failure, and non-generalizing local artifacts from 1.0 to 0.0 without missing any true mechanism, and backtracks to the correct state after mechanism reversal; and (d) admits no exploit-region evidence under proxy exploitation, raising held-out score by 0.209 at only +13.6% runtime over PiEvo .

In summary, our contributions are:

(a) We introduce evidence admission as the decision layer of principle-evolving agents: anomalies are attributed to latent sources, and revision is governed by asymmetric costs of false revision versus missed mechanism, replacing surprise thresholding as the trigger for changing scientific state.
(b) We propose Triage-PiEvo , which realizes this layer through per-channel trust estimation, a proposed–provisional–accepted–retracted principle lifecycle gated by independent held-out support, and an outcome-blind search portfolio, with all admission thresholds fixed before evaluation.
(c) We provide evidence separating search from revision: across four tasks and three backbones, Triage-PiEvo raises average discovery quality, and controlled corruption, hidden-source, reversal, and proxy-exploitation studies show that the gate rejects unreliable evidence without rejecting genuine mechanisms.

2 Related Work

2.1 Autonomous Scientific Agents and Evolving State

Scientific agents automate growing shares of the research cycle, from end-to-end paper production and multi-stage pipelines (Lu et al., 2024; Yamada et al., 2025; Tang et al., 2025; Schmidgall et al., 2025; Gottweis et al., 2025) to tool-coupled domain execution and closed-loop laboratory discovery (Bran et al., 2024; Boiko et al., 2023; Ruan et al., 2024; Yang et al., 2025; Ghafarollahi & Buehler, 2024; Ding et al., 2025; Ghareeb et al., 2026; Narayanan

et al., 2024; Laurent et al., 2024; Mitchener et al., 2025a; Skarlinski et al., 2024; Mitchener et al., 2025b). What persists between iterations there is an artifact, not a revisable scientific account. A narrower line makes the internal state editable: PiEvo and PiFlow expand a principle space on conflicting observations (Pu et al., 2026; 2025), ATLAS selects mechanism-discriminating experiments (Éltető et al., 2026), self-revising frameworks log regime changes (Wang & Buehler, 2026), and evolving architectures such as EvoMaster (Zhu et al., 2026) and ADMET-EvO (Zhou et al., 2026) pursue cumulative adaptation across tasks, with agentic laboratory world models unifying hypothesis generation and state tracking (Fu et al., 2026). Editable state is also corruptible: without explicit evidence admission, anomalous or corrupted feedback rewrites persistent principles directly (Ríos-García et al., 2026; Kirgis et al., 2026). Triage separates the history that steers search from the evidence authorized to alter scientific state.

2.2 Active Experimentation under Heterogeneous Sources

In active learning and Bayesian optimization, one line decides where to spend evaluations (MacKay, 1992; Houlsby et al., 2011), constraining exploration for safety (Sui et al., 2015) or scalability (Eriksson et al., 2019). Another exploits heterogeneous costs, fidelities, and failure rates across sources (Swersky et al., 2013; Kandasamy et al., 2017; Poloczek et al., 2017; Ghoreishi & Allaire, 2018), extending to unreliable channels (Mikkola et al., 2023; Fan et al., 2024), quality-controlled loops (Harris et al., 2025), and out-of-distribution expansion (De Santi et al., 2026). Both decide which observation to acquire before it is run, so an anomalous outcome inherits whatever reliability the acquisition rule assumed. The difference is in what the decision protects: multi-source methods weight a noisy source inside a regression model, while the object at risk here is a persistent principle that a single admitted artifact can corrupt for the rest of the run. Our AFE portfolio takes the acquisition-side view for candidate generation; Triage makes the post-observation decision of whether an anomaly may update that principle.

2.3 Verification, Provenance, and Agent Monitoring

In agent systems, verification targets transient outputs through feedback recycling (Yao et al., 2023; Shinn et al., 2023; Madaan et al., 2023), tool-based checking (Gou et al., 2023), independent verification questions (Dhuliawala et al., 2024), and evidence accumulation (Zhang et al., 2026), though self-correction without external evidence is unreliable (Huang et al., 2024; Kamoi et al., 2024). Related work argues for monitoring agentic systems before trusting them (Boston et al., 2026) and for immutable provenance trails over agent-generated artifacts (Liu et al., 2026), while Sibyl (Rangelov et al., 2026) uses temporal backtesting as held-out evidence. Triage carries these provenance and monitoring ideas to persistent principles: same-source agreement keeps observations search-active, acceptance requires independent held-out support, and contradictory held-out evidence forces retraction.

3 Methodology

3.1 Overview

Triage changes one interface in a principle-evolving agent: the path from an anomalous observation to a persistent principle update. A high-surprisal outcome opens a diagnostic state, in which the controller may repeat the evaluation, consult another channel, or request a held-out probe; only reconciled evidence can affect principle status.

Two linked histories support this. A primary record stores the candidate, prediction, outcome, source channel, and metadata. A verification record stores a repeated, cross-channel, or held-out outcome with its cost and a pointer to the primary observation. Separating them keeps a verification result from silently re-entering search as a new observation, and makes the provenance of each principle update explicit. We write Triage for this admission protocol and Triage-PiEvo for the integrated system of §4, which adds an outcome-blind Active

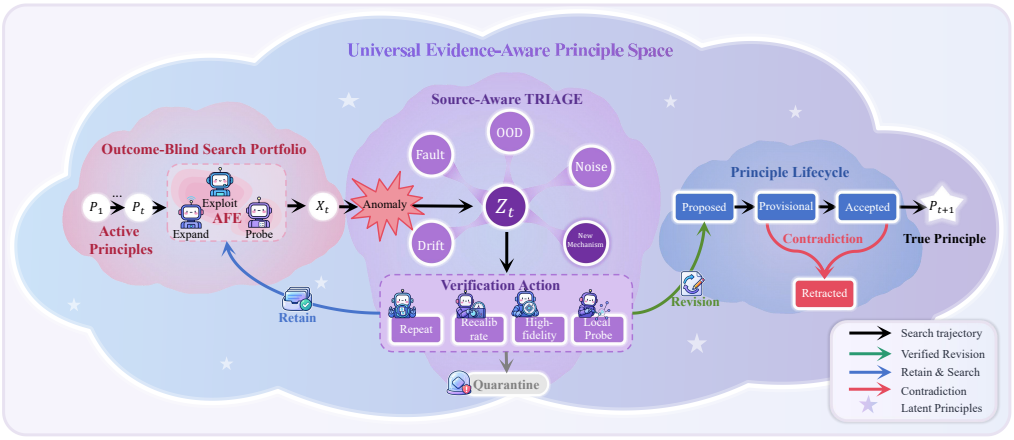


Figure 2: Overview of Triage-PiEvo. Left: an outcome-blind AFE portfolio cycles candidate proposals X_t across exploitation, frontier expansion, and mechanism-discriminating probes. Middle: Triage attributes each anomalous residual to a latent source Z_t and routes the observation to quarantine, search history, or principle revision. Right: principles advance Proposed → Provisional → Accepted only with independent held-out support; contradictory evidence retracts them.

Flow Expansion (AFE) portfolio. AFE decides where to search; Triage decides what may change a principle, so search stays exploratory without granting every promising observation the same epistemic standing (Figure 2).

3.2 Revision under Source Ambiguity

At step t, the agent evaluates a candidate x_t under principle state $\mathcal{P}_t$, obtaining outcome y_t with metadata m_t available before verification (batch identity, execution status, surrogate uncertainty); h_t is the history of primary observations and actions, and the anomaly score s_t summarizes the gap between y_t and the prediction implied by $\mathcal{P}_t$.

We represent the ambiguity behind this discrepancy with a latent source variable:

$$Z_t \in \mathcal{Z} = \{\text{noise},\ \text{device fault},\ \text{execution failure},\ \text{surrogate mismatch},\ \text{OOD input},\ \text{drift},\ \text{new mechanism}\}. \tag{1}$$

The true source is unobserved, and the same anomaly score can arise from several elements of $\mathcal{Z}$. A decision-theoretic account of this ambiguity starts from a posterior over sources,

$$q_t(z) = p(z \mid h_t, y_t, m_t, \mathcal{P}_t) \propto p(y_t \mid x_t, z, m_t, \mathcal{P}_t) p(z \mid h_t, m_t, \mathcal{P}_t). \tag{2}$$

Let $a_t \in \mathcal{A} = \{\text{repeat}, \text{recalibrate}, \text{high-fidelity}, \text{local probe}\}$ denote verification actions with cost $C(a)$. Deciding between revision and retention involves asymmetric errors:

$$L(d, z) = \ell_{\mathrm{FR}} \mathbb{1}[d = \textsf{revise}, z \neq z_{\mathrm{mech}}] + \ell_{\mathrm{FM}} \mathbb{1}[d = \textsf{retain}, z = z_{\mathrm{mech}}], \tag{3}$$

where $\ell_{\mathrm{FR}} \gg \ell_{\mathrm{FM}}$ penalizes corrupting persistent scientific principles much more heavily than missing a transient mechanism. The corresponding Bayes risk is $\mathcal{R}_t = \min_d \mathbb{E}_{q_t}[L(d, Z_t)]$, yielding the value-of-verification:

$$V_t(a) = \mathcal{R}_t - \mathbb{E}_{v \sim p(v \mid \mathcal{D}_t, a)}[\mathcal{R}_{t+1}(a, v)] - \lambda_c C(a) - \lambda_f Q^{\mathrm{FR}}_{t+1}(a). \tag{4}$$

From Reference Model to Runtime Protocol. Equations (1)–(4) state what a correct revision decision requires; they are not executed at runtime. Estimating $q_t(z)$ needs a per-source likelihood $p(y_t \mid x_t, z, m_t, \mathcal{P}_t)$ that these settings do not provide: the evaluator is a black box and source labels are never observed. The design question is thus not how to recover Z_t, but how to decide correctly without recovering it. We take three properties from the reference model and realize each with an observable quantity:

1. Ambiguity triggers verification, not revision. Where the reference model acts on posterior mass spread across $\mathcal{Z}$, the runtime opens a diagnostic state whenever $s_t > \tau_{\text{anom}} = 0.85$. Verification stays conditional, so cost is paid only on anomalies (2.8 extra calls per run)—the role of $\lambda_c C(a)$ in Equation (4).
2. Verification actions are ordered by discriminative power. Instead of maximizing $V_t(a)$ online, actions escalate along a fixed order—same-channel repeat, cross-channel check, held-out probe—stopping once the evidence is decisive. That order is the standing ranking of $V_t(a)$ under Equation (3): repeats separate transient noise from persistent effects, cross-channel checks separate channel faults from signal, and only held-out probes separate an artifact from a mechanism.
3. The gate is asymmetric because the losses are. With $\ell_{\text{FR}} \gg \ell_{\text{FM}}$, the two paths carry different burdens of proof. Same-channel agreement ($|y_t - v_t| \le \epsilon_{\text{rep}} = 0.12$) admits an observation to search but never authorizes revision; revision also needs agreement with a distinct channel $c' = \arg\max_{c'' \neq c} p^{\text{trust}}_{c''}$ ($|v^c_t - v^{c'}_t| \le \epsilon_{\text{ind}} = 0.12$) and k independent held-out records.

The controller tracks per-channel reliability with a Beta posterior $r_c \sim \text{Beta}(\alpha_c, \beta_c)$, $p^{\text{trust}}_c = \alpha_c/(\alpha_c + \beta_c)$, incrementing α_c on agreement and β_c on disagreement. Trust governs which channels are eligible to verify and flags degraded ones, but it never relaxes the standard of proof: k, τ_{gen}, and the tolerances are fixed a priori, so no channel earns the right to bypass held-out corroboration however reliable its history.

3.3 Outcome-Blind Search Portfolio

Triage-PiEvo broadens candidate proposals across three structured roles: best-region exploitation, frontier expansion, and mechanism-discriminating probes (De Santi et al., 2026).

Outcome-Blind Specification. "Outcome-blind" means the proposal policy is gate-blind and verifier-blind: generators cycle deterministically through the roles with no access to verifier channels, latent sources Z_t, or gate states, so a quarantine or retraction cannot steer the next proposal. Surrogate reward $\hat{\mu}(x)$ and uncertainty $\hat{\sigma}(x)$ are refit each round from admitted primary observations only—quarantined records are excluded until reconciled—so a corrupted outcome cannot influence scoring through the surrogate either. For role r,

$$A_r(x) = w^{\mu}_r \hat{\mu}(x) + w^{\sigma}_r \hat{\sigma}(x) + w^{n}_r N(x) + w^{d}_r D(x), \tag{5}$$

where $N(x)$ is distance from evaluated inputs and $D(x)$ descriptor novelty; role weights are fixed a priori (Appendix A) and Figure 3 shows the resulting geometry. Triage acts only after evaluation: agreement admits an observation to search, disagreement substitutes the verified median.

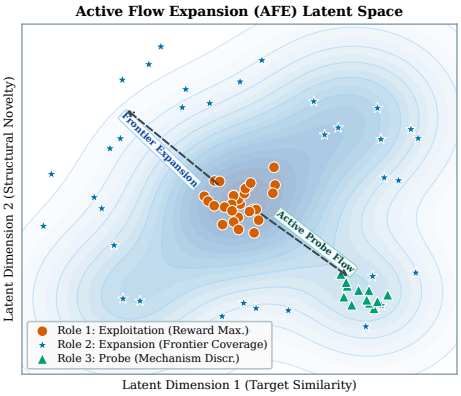


Figure 3: AFE portfolio (schematic). Complementary candidate roles.

3.4 Coherent Augmentation for Principles

Each candidate principle stores a parent pointer, supporting/contradicting records, and a lifecycle state: Proposed → Provisional → Accepted (or Retracted). The first supporting record makes a proposal provisional. A provisional principle is accepted if and only if

$$P : \text{provisional} \to \text{accepted} \quad \Longleftrightarrow \quad |\{e \in E(P) : e.\text{ind} \wedge e.\text{heldout} \wedge \text{support}(e, P)\}| \ge k, \tag{6}$$

with $k = 2$. Writing $\Delta_{\text{heldout}}(e; P)$ for the held-out gain of P measured by e, the three predicates are: e.ind, the record comes from an evaluator process sharing no memory, cache, or random seed with the primary evaluator—a declared channel identity checked before the record counts, asserted at the process level rather than tested statistically; e.heldout, the

record is evaluated on a partition X_{heldout} with $X_{\text{heldout}} \cap X_{\text{train}} = \emptyset$, fixed before the run and hidden from the generator; and stance, $\text{support}(e, P) = \mathbb{1}[\Delta_{\text{heldout}}(e; P) \geq \tau_{\text{gen}}]$ and $\text{contradict}(e, P) = \mathbb{1}[\Delta_{\text{heldout}}(e; P) \leq -\tau_{\text{gen}}]$ with $\tau_{\text{gen}} = 0.16$, leaving records inside the band uninformative. The symmetric margin is what separates an artifact from a mechanism: an artifact reproduces under repetition yet gives $\Delta_{\text{heldout}} < \tau_{\text{gen}}$, so it never reaches the count in Equation (6). Any record with $\text{contradict}(e, P) = 1$ retracts P and, recursively, every principle listing P as a parent. Pseudocode is in Appendix F.

4 Experiments

We evaluate Triage-PiEvo across three complementary dimensions: (1) downstream optimization performance across four scientific discovery tasks under three backbone LLMs; (2) component ablations isolating search expansion from evidence admission; and (3) controlled stress tests targeting evaluator corruption, hidden anomaly sources, mechanism reversal, proxy exploitation, and cross-harness deployment.

4.1 Experiment Setup

Tasks and Baselines. We use four discovery environments from PiEvo (Pu et al., 2026): Molecular Bio-activity Optimization (MBO), Nanomaterial Optical Property Optimization (NHO), Superconductor Optimization (SPO), and Transition Metal Complex Optimization (TMC), against seven baselines: Vanilla MAS, ReAct (Yao et al., 2023), AI Scientist v1 (Lu et al., 2024), AI Scientist v2 (Yamada et al., 2025), AI Researcher (Tang et al., 2025), PiFlow (Pu et al., 2025), and PiEvo (Pu et al., 2026).

Implementation Details. We evaluate three backbones—Qwen3-235B, Gemini-2.5-Flash-Lite, GPT-5.6-Luna—in direct execution mode without extended chain-of-thought, over three fixed seeds (0, 1, 2) at a budget of 24 primary evaluations: 96 runs per model, 288 trajectories. Gate parameters are fixed a priori across all tasks and backbones: $\tau_{\text{anom}} = 0.85$, $\epsilon_{\text{rep}} = \epsilon_{\text{ind}} = 0.12$, $\tau_{\text{gen}} = 0.16$, $k = 2$ (full specification and sensitivity sweeps in Appendix A).

Evaluation Metrics. With $y_{1:T}$ the primary outcomes of a run: (1) Solution Quality (SQ, %, ↑) is $100 \cdot \max_t y_t / \mu_{\text{ref}}$; (2) AUOC (%, ↑) is $100 \cdot \frac{1}{T} \sum_t \max_{t' \leq t} y_{t'} / \mu_{\text{ref}}$, crediting how early quality is reached; (3) APD (↑) is mean pairwise cosine distance between proposed candidates under Qwen3-Embedding-0.6B. References are $\mu_{\text{ref}} = 6.5$ (MBO), 2.0 (NHO), 298.5 (SPO), 493.8 (TMC), from PiEvo (Pu et al., 2026). These are published reference values, not attainable upper bounds, so SQ passes 100 when a run beats the reference; we apply no clipping.

4.2 Matched Downstream Performance

Table 1 reports terminal Solution Quality. Triage-PiEvo reaches the highest average SQ for each backbone—82.10, 79.63, and 83.37—improving on the strongest baseline for that backbone by 11.75, 11.14, and 7.48 points. That strongest baseline differs across backbones (PiEvo , PiFlow, ReAct), and none leads on more than one: a method tuned to one backbone's failure profile does not carry to another, whereas Triage-PiEvo is first on all three.

The per-task breakdown locates the gain. It concentrates on MBO (132.00, 115.00, 136.00), the task with the widest baseline spread—on Qwen3-235B, from 73.37 to 105.70—and therefore the most room for a principle to be either genuinely revised or corrupted by a spurious outcome. SPO is the opposite: every method lands between 30.74 and 42.00 and our margin narrows to 0.6–3.4 points. Gains track how much of a task's difficulty lies in deciding what to believe rather than where to search, which is what an evidence-admission layer should predict. Variance moves the same way: on Qwen3-235B MBO, Triage-PiEvo reports ±9.84 against ±27.49 for the closest baseline and ±39.82 for the widest, and on all three backbones our MBO spread stays below the baseline median. Since the gate keeps unreconciled

Table 1: Solution Quality (SQ, %) across four discovery tasks and three backbone LLMs. Entries report mean ± SD over three seeds; averages are computed across tasks. Bold indicates the best result within each backbone.

Model / Method	MBO	NHO	SPO	TMC	Average
Qwen3-235B (No Thinking)					
Vanilla MAS	83.35 ± 29.95	52.83 ± 20.36	37.29 ± 0.75	56.34 ± 3.69	57.45
ReAct	75.48 ± 4.94 ↓ 0.9x	61.95 ± 17.08 ↑ 1.2x	37.49 ± 0.46	72.80 ± 4.48 ↑ 1.3x	61.93 ↑ 1.1x
AI Scientist v1	73.37 ± 39.82 ↓ 0.9x	41.50 ± 3.12 ↓ 0.8x	37.04 ± 0.61	63.96 ± 8.93 ↑ 1.1x	53.97 ↓ 0.9x
AI Scientist v2	84.40 ± 20.07	52.34 ± 15.56	41.40 ± 1.15 ↑ 1.1x	63.52 ± 2.86 ↑ 1.1x	60.42 ↑ 1.1x
AI Researcher	105.70 ± 21.92 ↑ 1.3x	37.21 ± 5.68 ↓ 0.7x	37.39 ± 0.85	78.80 ± 3.76 ↑ 1.4x	64.78 ↑ 1.1x
PiFlow	81.89 ± 30.49	56.15 ± 15.23 ↑ 1.1x	37.21 ± 0.52	54.58 ± 2.70	57.46
PiEvo	97.84 ± 27.49 ↑ 1.2x	65.53 ± 21.90 ↑ 1.2x	37.71 ± 0.53	80.30 ± 4.57 ↑ 1.4x	70.35 ↑ 1.2x
Triage-PiEvo (ours)	132.00 ± 9.84 ↑ 1.6x	72.89 ± 17.80 ↑ 1.4x	42.00 ± 1.20 ↑ 1.1x	81.50 ± 4.02 ↑ 1.4x	82.10 ↑ 1.4x
Gemini-2.5-Flash-Lite (No Thinking)					
Vanilla MAS	62.06 ± 14.09	71.98 ± 1.01	36.95 ± 0.46	66.64 ± 17.42	59.41
ReAct	58.83 ± 6.44 ↓ 0.9x	69.90 ± 1.50	32.06 ± 5.75 ↓ 0.9x	56.31 ± 2.37 ↓ 0.8x	54.27 ↓ 0.9x
AI Scientist v1	75.60 ± 22.03 ↑ 1.2x	73.58 ± 2.23	37.86 ± 2.55	55.70 ± 2.62 ↓ 0.8x	60.68
AI Scientist v2	99.30 ± 22.73 ↑ 1.6x	74.87 ± 2.00	30.74 ± 3.96 ↓ 0.8x	54.80 ± 4.09 ↓ 0.8x	64.93 ↑ 1.1x
AI Researcher	66.47 ± 16.70 ↑ 1.1x	65.39 ± 9.16 ↓ 0.9x	31.62 ± 3.82 ↓ 0.9x	63.46 ± 13.85	56.74
PiFlow	96.59 ± 11.71 ↑ 1.6x	74.24 ± 4.06	34.91 ± 3.66 ↓ 0.9x	68.22 ± 8.71	68.49 ↑ 1.2x
PiEvo	72.59 ± 6.70 ↑ 1.2x	81.00 ± 3.06 ↑ 1.1x	38.02 ± 0.50	78.50 ± 7.60 ↑ 1.2x	67.53 ↑ 1.1x
Triage-PiEvo (ours)	115.00 ± 10.00 ↑ 1.9x	83.00 ± 5.00 ↑ 1.2x	40.50 ± 4.00 ↑ 1.1x	80.00 ± 4.00 ↑ 1.2x	79.63 ↑ 1.3x
GPT-5.6-Luna (No Thinking)					
Vanilla MAS	91.84 ± 18.08	66.26 ± 7.14	36.42 ± 0.50	68.07 ± 3.91	65.65
ReAct	132.00 ± 22.98 ↑ 1.4x	69.98 ± 0.14 ↑ 1.1x	36.22 ± 0.34	65.37 ± 5.39	75.89 ↑ 1.2x
AI Scientist v1	91.16 ± 19.35	53.02 ± 12.32 ↓ 0.8x	35.52 ± 4.90	62.44 ± 6.24 ↓ 0.9x	60.54 ↓ 0.9x
AI Scientist v2	89.63 ± 31.45	65.13 ± 15.46	36.77 ± 5.05	58.29 ± 0.95 ↓ 0.9x	62.46
AI Researcher	77.86 ± 15.38 ↓ 0.8x	73.92 ± 9.01 ↑ 1.1x	36.28 ± 0.24	72.93 ± 2.06 ↑ 1.1x	65.25
PiFlow	128.77 ± 18.74 ↑ 1.4x	46.62 ± 17.77 ↓ 0.7x	31.49 ± 5.94 ↓ 0.9x	59.59 ± 5.23 ↓ 0.9x	66.62
PiEvo	69.75 ± 6.20 ↓ 0.8x	67.82 ± 10.87	37.41 ± 0.53	76.50 ± 1.78 ↑ 1.1x	62.87
Triage-PiEvo (ours)	136.00 ± 9.84 ↑ 1.5x	78.18 ± 5.44 ↑ 1.2x	40.80 ± 5.70 ↑ 1.1x	78.50 ± 3.75 ↑ 1.2x	83.37 ↑ 1.3x

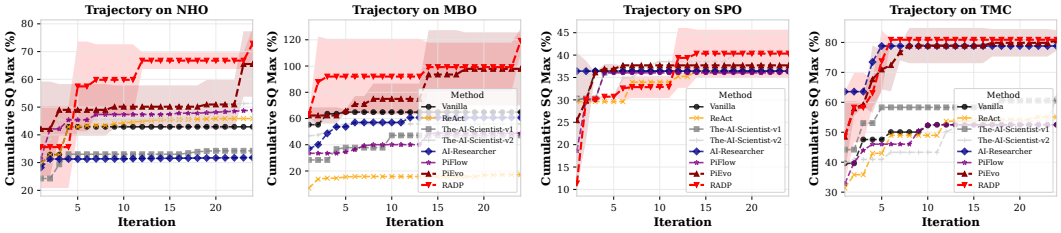


Figure 4: Best-so-far optimization trajectories over 24 primary evaluations across four benchmark discovery tasks with Qwen3-235B. Curves report the empirical mean over three independent seeds; shaded ribbons indicate ±1 standard deviation. Across all scientific domains, Triage-PiEvo rapidly accumulates high-quality candidates and sustains the highest terminal solution quality.

outcomes out of persistent state, suppressing seed-to-seed spread is a more direct signature of it working than the mean shift is.

This table evaluates the integrated system and does not attribute the gain to either component; §4.3 separates them. AUOC and APD appear in Table 2, where higher peak SQ coincides with broader coverage and faster optimization—the gate does not buy reliability by narrowing exploration. Figure 4 shows best-so-far trajectories with Qwen3-235B: Triage-PiEvo opens a margin within 5 iterations and holds it to the terminal rounds, at +13.6% runtime (Appendix C).

4.3 Component Study: Search Expansion vs. Evidence Admission

To separate candidate generation from evidence triage, Table 3 reports a 48-cell factorial ablation on Qwen3-235B at budget 8. The components fail in opposite directions. Triage-only falls below PiEvo on clean SQ (60.60% vs. 64.78%): without search expansion, filtering only narrows an already narrow candidate flow, and clean tasks give it little to filter. AFE

Table 2: Average Pairwise Distance (APD) and Area Under the Optimisation Curve (AUOC) across four tasks and three backbone LLMs. Entries report mean ± SD over three seeds. Higher APD indicates broader candidate coverage, while higher AUOC indicates faster accumulation of high-quality candidates.

Model / Method	MBO		NHO		SPO		TMC		Average	
	APD ↑	AUOC ↑	APD ↑	AUOC ↑	APD ↑	AUOC ↑	APD ↑	AUOC ↑	Avg APD ↑	Avg AUOC ↑
Qwen3-235B (No Thinking)										
Vanilla MAS	53.58±15.25	70.09±17.75	32.23±3.72	49.33±17.82	52.03±1.05	36.34±0.38	51.89±3.85	52.67±3.49	47.43	52.11
ReAct	59.43±8.60	70.58±7.95	33.31±1.86	47.46±7.04	56.13±4.49	35.74±0.76	72.93±0.33	66.95±4.67	55.45 ↑ 1.2x	55.18 ↑ 1.1x
AI Scientist v1	52.93±15.22	56.16±26.05	32.98±2.32	38.84±2.06	42.55±5.61	35.62±1.46	64.43±7.02	57.50±6.15	48.22	47.03 ↓ 0.9x
AI Scientist v2	60.26±10.57	76.03±7.92	28.39±3.49	39.51±3.29	48.18±1.41	37.03±0.22	55.66±2.28	54.75±6.66	48.12	51.83
AI Researcher	59.17±3.77	89.21±19.08	25.51±3.45	34.51±3.80	47.62±3.55	36.82±0.41	52.46±6.03	74.19±0.70	46.19	58.68 ↑ 1.1x
PiFlow	64.99±2.39	64.04±14.87	28.07±1.85	52.93±15.82	51.24±4.14	35.90±1.33	57.76±3.35	50.26±1.85	50.52 ↑ 1.1x	50.78
PiEvo	63.41±10.78	84.14±16.22	23.80±5.23	52.50±13.36	54.41±1.42	36.83±0.86	71.76±2.58	74.93±5.05	53.35 ↑ 1.1x	62.10 ↑ 1.2x
Triage-PiEvo (ours)	73.71±2.86	110.00±18.41	23.67±0.79	61.21±11.52	57.34±0.26	38.50±1.80	74.20±1.41	77.20±2.95	57.23 ↑ 1.2x	71.73 ↑ 1.4x
Gemini-2.5-Flash-Lite (No Thinking)										
Vanilla MAS	65.06±5.46	46.01±15.24	33.55±0.90	61.71±2.41	53.65±1.10	35.45±0.05	57.23±5.29	62.88±15.97	52.37	51.51
ReAct	62.08±1.15	44.54±2.54	34.46±0.57	58.08±5.33	42.17±12.13	25.95±8.19	52.75±5.72	52.03±0.12	47.87 ↓ 0.9x	45.15 ↓ 0.9x
AI Scientist v1	52.02±15.45	58.73±5.53	33.13±0.89	64.99±5.57	47.13±4.98	33.96±1.93	56.53±1.71	51.60±0.09	47.20 ↓ 0.9x	52.32
AI Scientist v2	46.27±12.65	59.35±11.81	35.09±0.38	67.18±1.82	43.28±0.12	24.33±7.17	42.16±9.83	52.77±3.83	41.70 ↓ 0.8x	50.91
AI Researcher	60.38±7.64	45.98±15.08	33.16±3.51	58.94±6.79	41.65±9.04	28.78±2.15	55.97±2.51	59.77±14.66	47.79 ↓ 0.9x	48.37 ↓ 0.9x
PiFlow	62.64±6.57	68.05±8.07	29.73±4.14	65.72±5.25	44.03±2.95	33.27±2.48	66.17±6.06	63.89±6.66	50.64	57.73 ↑ 1.1x
PiEvo	64.40±1.50	71.40±7.88	23.38±0.37	65.97±3.31	53.88±0.59	35.02±1.92	70.35±0.76	69.74±7.30	53.00	60.53 ↑ 1.2x
Triage-PiEvo (ours)	70.00±5.00	108.00±12.00	24.00±2.00	76.00±8.00	57.00±1.50	36.00±4.00	72.50±2.50	75.00±6.00	55.88 ↑ 1.1x	73.75 ↑ 1.4x
GPT-5.6-Luna (No Thinking)										
Vanilla MAS	64.66±4.12	69.17±12.39	27.30±1.90	60.18±3.66	48.73±2.51	34.43±0.60	71.55±1.07	63.26±4.82	53.06	56.76
ReAct	38.36±20.70	86.71±13.76	31.58±2.81	59.57±2.38	49.26±2.90	33.88±0.86	66.05±7.21	62.28±5.12	46.31 ↓ 0.9x	60.61 ↑ 1.1x
AI Scientist v1	55.57±18.24	71.58±4.38	29.17±3.01	44.85±6.39	47.10±0.71	29.87±4.50	53.91±3.25	54.73±0.57	46.44 ↓ 0.9x	50.26 ↓ 0.9x
AI Scientist v2	51.59±19.99	60.12±11.92	24.07±6.68	54.21±13.33	45.21±3.80	32.19±5.66	49.34±11.99	51.65±1.98	42.55 ↓ 0.8x	49.54 ↓ 0.9x
AI Researcher	66.12±1.10	70.56±16.40	27.41±4.25	65.81±4.10	50.05±3.05	32.56±1.91	71.88±0.71	65.59±2.27	53.86	58.63
PiFlow	48.90±24.99	79.51±4.35	17.19±2.39	38.66±10.82	37.93±7.95	28.15±7.38	56.56±3.30	56.09±6.93	40.14 ↓ 0.8x	50.60 ↓ 0.9x
PiEvo	64.05±1.98	62.95±1.52	23.83±0.49	52.04±7.69	53.07±0.84	36.29±1.21	73.20±0.54	74.81±0.97	53.54	56.52
Triage-PiEvo (ours)	73.72±2.84	104.00±19.33	24.08±1.10	68.77±8.52	57.87±0.88	38.10±1.60	74.80±1.20	76.80±3.20	57.62 ↑ 1.1x	71.92 ↑ 1.3x

without the gate restores exploration but admits whatever the evaluator reports. Only the combination leads on both metrics (65.98% SQ, 60.45% AUOC). AFE sets where the agent looks, Triage sets what may change persistent state—which is why the gate costs a little quality on clean tasks and repays it when feedback is unreliable (§4.4).

Table 3: Qwen3-235B component study across four tasks at budget 8. SQ and AUOC are averaged across tasks and seeds.

Variant	Average SQ ↑	Average AUOC ↑	Full cells
PiEvo	64.78	58.15	11/12
Triage-only	60.60	55.08	11/12
Triage-PiEvo w/o Gate	62.73	58.23	12/12
Full Triage-PiEvo	65.98	60.45	12/12

4.4 Stress Tests: Corruption, Hidden Sources, and Proxy Exploitation

Evaluator Corruption and Hidden Sources. We first corrupt the evaluator directly, running Qwen3-235B on MBO under clean conditions, transient perturbations $(+1.0, +2.5)$, and a persistent $+1.5$ bias across three consecutive calls (48 cells). Without the gate every corrupted observation enters the search history (1.000 ± 0.000); with it, none does (0.000 ± 0.000), each replaced by the verified median. We then hide the source: 144 trials span measurement noise, surrogate mismatch, baseline drift, a true hidden mechanism, local non-generalizing artifacts, and mechanism reversal. A source-blind updater expands its principle set on noise, surrogate mismatch, and local artifacts in every trial (false-expansion rate 1.0); Triage reaches 0.0 on all artifact conditions while recovering the true mechanism with 1.0 final-state accuracy. The held-out test does this work, not repetition: local artifacts reproduce perfectly, so only their failure to transfer sets them apart from a mechanism. These interventions corrupt the primary channel while leaving verification intact; Appendix A reports how the protective effect attenuates as the two channels fail together.

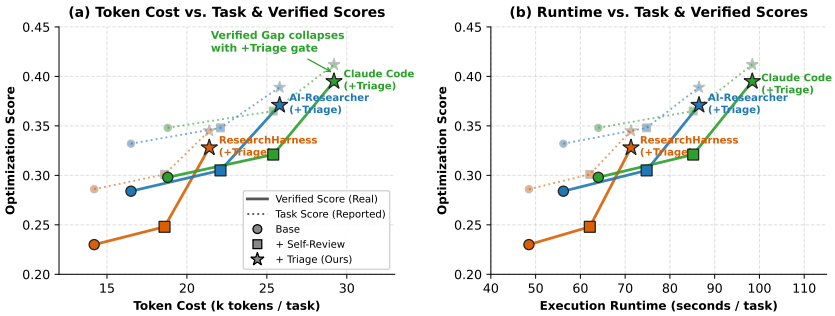


Figure 5: Cost-efficiency trade-offs across autonomous research harnesses. (a) Token expenditure versus reported Task Score and externally Verified Score. Triage substantially narrows the verification gap with minimal token overhead. (b) Wall-clock execution runtime versus optimization performance across ResearchHarness, AI-Researcher, and Claude Code. Full numerical results are reported in Table 6 (Appendix D).

Retraction after Mechanism Reversal. When the mechanism reverses mid-trajectory, an accepted principle becomes actively misleading. In a materials trajectory, an environmental shift at $t = 16$ inverts the optimal valence state; PiEvo stays anchored to $\mathcal{P}$ and stagnates. Triage-PiEvo flags an anomalous residual at $t = 17$, issues held-out probes, retracts $\mathcal{P}$ with its descendants at $t = 18$ once held-out evidence contradicts it, and recovers the new optimum by $t = 23$. Retraction turns the gate from a filter into a revisable commitment: the criterion that admits a principle also removes it.

Proxy Exploitation. A gate that only checks reproducibility stays exposed to reward hacking, because an exploit reproduces perfectly. Under an action-dependent proxy mismatch where $x \geq 0.72$ inflates apparent reward while true reward peaks at $x = 0.32$, ungated variants admit the exploit every round (5.33 ± 0.58 false admissions for PiEvo, 6.00 ± 0.00 for Triage-PiEvo w/o Gate). Triage-PiEvo quarantines them, records none, raises held-out score by $+0.209 \pm 0.054$, and cuts true regret by 1.252 ± 0.323. Full numbers for this and the harness study are in Tables 5–7.

Cross-Harness Portability and Cost. We attach Triage to ResearchHarness, AI-Researcher (Tang et al., 2025), and Claude Code. Each shows a gap between reported Task Score and externally Verified Score traceable to unverified intermediates—malformed SMILES and simulator boundary artifacts read as discoveries (Figure 5). Triage narrows it by 0.087–0.098, lowers critical failures from 14.8%–26.2% to 5.2%, and cuts false admissions by over 60%, at 2.8 verification calls per run (+$0.24) and +13.0%–+16.7% token overhead, with no non-terminating run (Appendices D–E).

5 Conclusion

An anomalous residual is source-ambiguous: a genuine mechanism, a transient evaluator error, a surrogate misspecification, and a proxy exploit can all produce the same surprisal. Surprise-driven agents act on it directly, so whatever is unreliable in it propagates into persistent state. We introduced Triage, which separates exploratory search from principle revision and holds a principle provisional until independent held-out records support it, retracting it and its descendants when held-out evidence contradicts it. Paired with outcome-blind search expansion in Triage-PiEvo, this gives the highest average solution quality in our matched setup across four domains and three backbones, admits no corrupted evidence under the interventions we construct, rejects the tested artifacts while retaining the tested mechanism, and recovers after mechanism reversal; the ablation locates the effect in the held-out criterion rather than in repetition or channel bookkeeping. Evidence good enough to direct the next experiment and evidence good enough to rewrite what an agent believes are different standards, and an agent that conflates them will eventually believe its own noise.

AI use statement

In this research, generative AI tools were employed for coding assistance in data visualization scripts and for LaTeX drafting/grammar editing. All scientific concepts, decision-theoretic formulations, algorithmic designs, experimental benchmarks, empirical results, and analytical conclusions were conceived, executed, verified, and audited by the authors, who take full responsibility for the integrity and content of this paper.

Ethics statement

This work develops automated methods for scientific discovery and evidence verification. The evaluated benchmarks (molecular optimization, optical properties, superconductors, transition metal complexes) rely on publicly available scientific datasets and simulation proxies. No human subjects, animal trials, or hazardous materials were involved. Potential dual-use considerations of automated discovery agents are addressed through our explicit evidence-quarantine mechanisms, which prevent unverified or anomalous outputs from automatically altering operational protocols.

Reproducibility statement

To ensure complete reproducibility, all algorithm parameters, admission thresholds, and experimental budgets are specified in Section 4 and Appendix A. The evaluation harness, seed configurations (seeds 0–2 for main benchmarks), evaluator interfaces, and verification protocols are documented in Section 4 and Appendices A–G. Anonymous code and run manifests will be made available upon publication.

References

Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. Nature, 624:570–578, 2023. doi: 10.1038/s41586-023-06792-0.

Samuel Boston, Harrison Clark, Patrick Davis, et al. Monitoring agentic systems before they're reliable. arXiv preprint arXiv:2604.07823, 2026.

Andres M. Bran, Sam Cox, Oliver Schilter, Carlo Baldassari, Andrew D. White, and Philippe Schwaller. ChemCrow: Augmenting large-language models with chemistry tools. Nature Machine Intelligence, 6:525–535, 2024. doi: 10.1038/s42256-024-00832-8.

Riccardo De Santi, Bruce Lee, Cristian Perez Jensen, Kimon Protopapas, Sophia Tang, Cheng-Hao Liu, Pranam Chatterjee, Yisong Yue, and Andreas Krause. Active flow expansion for out-of-distribution discovery: from theory to molecules, 2026. URL https://arxiv.org/abs/2606.08802.

Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. Chain-of-verification reduces hallucination in large language models. In Findings of the Association for Computational Linguistics: ACL 2024, pp. 3563–3578, 2024. doi: 10.18653/v1/2024.findings-acl.212.

Keyan Ding, Jing Yu, Junjie Huang, Yuchen Yang, Qiang Zhang, and Huajun Chen. SciToolAgent: A knowledge graph-driven scientific agent for multi-tool integration. Nature Computational Science, 5:962–972, 2025. doi: 10.1038/s43588-025-00849-y.

Noémi Éltető, Nathaniel D. Daw, Kimberly L. Stachenfeld, and Kevin J. Miller. ATLAS: Active theory learning for automated science, 2026. URL https://arxiv.org/abs/2606.12386.

David Eriksson, Michael Pearce, Jacob R. Gardner, Ryan Turner, and Matthias Poloczek. Scalable global optimization via local bayesian optimization. In Advances in Neural Information Processing Systems, volume 32, 2019.

Mingzhou Fan, Byung-Jun Yoon, Edward Dougherty, Nathan Urban, Francis Alexander, Raymundo Arróyave, and Xiaoning Qian. Multi-fidelity bayesian optimization with multiple information sources of input-dependent fidelity. In Proceedings of the Fortieth Conference on Uncertainty in Artificial Intelligence, volume 244 of Proceedings of Machine Learning Research, pp. 1271–1293. PMLR, 2024. URL https://proceedings.mlr.press/v244/fan24a.html.

Yanlin Fei, Nazhou Liu, Xinmiao Yu, Shaolong Chen, Lei Li, Rahul Thapa, Madalina Ciobanu, Navan Preet Singh, Qingqing Mao, and Ritankar Das. How do agents fail on AutoResearch: End-to-end diagnostic evaluation on 100 real-world frontier research tasks, 2026. URL https://arxiv.org/abs/2608.14905.

Xiang Fu, Ming Yang, Zhi Sun, et al. Agentic laboratories of the future: Towards world models for scientific discovery. arXiv preprint arXiv:2603.01128, 2026.

Alireza Ghafarollahi and Markus J. Buehler. Sciagents: Automating scientific discovery through multi-agent intelligent graph reasoning, 2024. URL https://arxiv.org/abs/2409.05556.

Ali E. Ghareeb, Benjamin Chang, Ludovico Mitchener, Angela Yiu, Caralyn J. Szostkiewicz, Dmytro Shved, Gavin J. Gyimesi, Jon M. Laurent, Samantha M. Wright, Muhammed T. Razzak, et al. A multi-agent system for automating scientific discovery. Nature, 655: 497–505, 2026. doi: 10.1038/s41586-026-10652-y.

Seyede Fatemeh Ghoreishi and Douglas Allaire. Multi-information source constrained bayesian optimization. Structural and Multidisciplinary Optimization, 59(3):977–991, 2018. doi: 10.1007/s00158-018-2115-z.

Juraj Gottweis, Wei-Hung Weng, Alexander Daryin, Tao Tu, Anil Palepu, Petar Sirkovic, Artiom Myaskovsky, Felix Weissenberger, Keran Rong, Ryutaro Tanno, et al. Towards an ai co-scientist, 2025. URL https://arxiv.org/abs/2502.18864.

Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. Critic: Large language models can self-correct with tool-interactive critiquing, 2023. URL https://arxiv.org/abs/2305.11738.

Sumner B. Harris, Rama Vasudevan, and Yongtao Liu. Active oversight and quality control in standard bayesian optimization for autonomous experiments. npj Computational Materials, 11:23, 2025. doi: 10.1038/s41524-024-01485-2.

Neil Houlsby, Ferenc Huszár, Zoubin Ghahramani, and Máté Lengyel. Bayesian active learning for classification and preference learning, 2011. URL https://arxiv.org/abs/1112.5745.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large language models cannot self-correct reasoning yet. In International Conference on Learning Representations, 2024. URL https://openreview.net/forum?id=IkmD3fKBPQ.

Ryo Kamoi, Yusen Zhang, Nan Zhang, Jiawei Han, and Rui Zhang. When can LLMs actually correct their own mistakes? a critical survey of self-correction of LLMs. Transactions of the Association for Computational Linguistics, 12:1417–1440, 2024. doi: 10.1162/tacl_a_00713.

Kirthevasan Kandasamy, Gautam Dasarathy, Jeff Schneider, and Barnabás Póczos. Multi-fidelity bayesian optimisation with continuous approximations. In Proceedings of the 34th International Conference on Machine Learning, volume 70 of Proceedings of Machine Learning Research, pp. 1799–1808, 2017. URL https://proceedings.mlr.press/v70/kandasamy17a.html.

Peter Kirgis, Sayash Kapoor, Andrew Schwartz, Stephan Rabanser, David Africa, Konstantinos Voudouris, Viet Nguyen, Toby Pilditch, Magda Dubois, Harry Coppock, Cozmin

Ududec, Nitya Nadgir, Matilda Orona, Tilman Bayer, Derrick Chan-Sew, Yue Ling, Abhishek Shetty, Helen Toner, Gillian Hadfield, Seth Lazar, Steve Newman, Shoshannah Tekofsky, Rishi Bommasani, and Arvind Narayanan. Can ai agents conduct open-ended ai research? early evidence from two case studies, 2026. URL https://arxiv.org/abs/2607.27191.

Jon M. Laurent, Michal Janik, Michael H. Lee, Kevin L. Grayson, Gabriel Ferrand, Gabriel J. Rocklin, et al. Lab-bench: Measuring capabilities of language models for biology research, 2024. URL https://arxiv.org/abs/2407.10362.

Eric Liu, Sarah Mitchell, Kevin Tan, et al. The last human-written paper: Agent-native research artifacts. arXiv preprint arXiv:2605.02341, 2026.

Chris Lu, Cong Lu, Robert Tjarko Lange, Jakob Foerster, Jeff Clune, and David Ha. The ai scientist: Towards fully automated open-ended scientific discovery, 2024. URL https://arxiv.org/abs/2408.06292.

David J. C. MacKay. Information-based objective functions for active data selection. Neural Computation, 4(4):590–604, 1992. doi: 10.1162/neco.1992.4.4.590.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative refinement with self-feedback. In Advances in Neural Information Processing Systems, volume 36, 2023. doi: 10.52202/075280-2019.

Petrus Mikkola, Julien Martinelli, Louis Filstroff, and Samuel Kaski. Multi-fidelity bayesian optimization with unreliable information sources. In Proceedings of the 26th International Conference on Artificial Intelligence and Statistics, volume 206 of Proceedings of Machine Learning Research, pp. 7425–7454. PMLR, 2023. URL https://proceedings.mlr.press/v206/mikkola23a.html.

Ludovico Mitchener, Jon M. Laurent, Samuel G. Rodriques, Andrew D. White, et al. Bixbench: A comprehensive benchmark for llm-based agents in computational biology, 2025a. URL https://arxiv.org/abs/2503.00096.

Ludovico Mitchener, Jon M. Laurent, Samuel G. Rodriques, Andrew D. White, et al. Kosmos: An ai scientist for autonomous discovery, 2025b. URL https://arxiv.org/abs/2511.02824.

Siddharth Narayanan, James D. Braza, Ryan-Rhys Griffiths, Jon M. Laurent, Sam Cox, Samuel G. Rodriques, Markus J. Buehler, Andrew D. White, et al. Aviary: training language agents on challenging scientific tasks, 2024. URL https://arxiv.org/abs/2412.21154.

Matthias Poloczek, Jialei Wang, and Peter I. Frazier. Multi-information source optimization. In Advances in Neural Information Processing Systems, volume 30, 2017.

Yingming Pu, Tao Lin, and Hongyu Chen. Piflow: Principle-aware scientific discovery with multi-agent collaboration, 2025. URL https://arxiv.org/abs/2505.15047.

Yingming Pu, Tao Lin, and Hongyu Chen. Principle-evolvable scientific discovery via uncertainty minimization. In Proceedings of the 43rd International Conference on Machine Learning, volume 306 of Proceedings of Machine Learning Research, 2026. URL https://arxiv.org/abs/2602.06448.

Dimitar Rangelov, Bojan Ivanov, Jennifer Smith, et al. Sibyl: Temporal backtesting for literature-based scientific discovery with large language model agents. arXiv preprint arXiv:2606.09145, 2026.

Martiño Ríos-García, Nawaf Alampara, Chandan Gupta, Indrajeet Mandal, Sajid Mannan, Ali Asghar Aghajani, N. M. Anoop Krishnan, and Kevin Maik Jablonka. Ai scientists produce results without reasoning scientifically, 2026. URL https://arxiv.org/abs/2604.18805.

Yixiang Ruan, Chenyin Lu, Ning Xu, Yuchen He, Yixin Chen, Jian Zhang, Jun Xuan, Jianzhang Pan, Qun Fang, Hanyu Gao, Xiaodong Shen, Ning Ye, Qiang Zhang, and Yiming Mo. An automatic end-to-end chemical synthesis development platform powered by large language models. Nature Communications, 15:10160, 2024. doi: 10.1038/s41467-024-54457-x.

Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael Moor, Zicheng Liu, and Emad Barsoum. Agent laboratory: Using LLM agents as research assistants. In Findings of the Association for Computational Linguistics: EMNLP 2025, pp. 5977–6043. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-emnlp.320.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In Advances in Neural Information Processing Systems, volume 36, 2023. doi: 10.52202/075280-0377.

Michael D. Skarlinski, Sam Cox, Jon M. Laurent, James D. Braza, Michaela M. Hinks, Michael J. Hammerling, Manvitha Ponnapati, Samuel G. Rodriques, and Andrew D. White. Language agents achieve superhuman synthesis of scientific knowledge, 2024. URL https://arxiv.org/abs/2409.13740.

Yanan Sui, Alkis Gotovos, Joel Burdick, and Andreas Krause. Safe exploration for optimization with gaussian processes. In Proceedings of the 32nd International Conference on Machine Learning, volume 37 of Proceedings of Machine Learning Research, pp. 997–1005, 2015. URL https://proceedings.mlr.press/v37/sui15.html.

Kevin Swersky, Jasper Snoek, and Ryan P. Adams. Multi-task bayesian optimization. In Advances in Neural Information Processing Systems, volume 26, 2013.

Jiabin Tang, Lianghao Xia, Zhonghang Li, and Chao Huang. AI-researcher: Autonomous scientific innovation. In Advances in Neural Information Processing Systems, volume 38, 2025. doi: 10.52202/085713-0320.

Fiona Y. Wang and Markus J. Buehler. Self-revising discovery systems for science: A categorical framework for agentic artificial intelligence, 2026. URL https://arxiv.org/abs/2606.01444.

Yutaro Yamada, Robert Tjarko Lange, Cong Lu, Shengran Hu, Chris Lu, Jakob Foerster, Jeff Clune, and David Ha. The ai scientist-v2: Workshop-level automated scientific discovery via agentic tree search, 2025. URL https://arxiv.org/abs/2504.08066.

Zonglin Yang, Wanhao Liu, Ben Gao, Tong Xie, Yuqiang Li, Wanli Ouyang, Soujanya Poria, Erik Cambria, and Dongzhan Zhou. MOOSE-Chem: Large language models for rediscovering unseen chemistry scientific hypotheses. In International Conference on Learning Representations, 2025.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing reasoning and acting in language models. In International Conference on Learning Representations, 2023. URL https://openreview.net/forum?id=WE_vluYUL-X.

Yichi Zhang, Nabeel Seedat, Yinpeng Dong, Peng Cui, Jun Zhu, and Mihaela van der Schaar. Guideline-grounded evidence accumulation for high-stakes agent verification. In International Conference on Machine Learning, 2026. URL https://arxiv.org/abs/2603.02798.

Tian Zhou, Meng Zhang, Wei Liu, et al. Admet-evo: a self-evolving scientific agent for sustained research across heterogeneous tasks. arXiv preprint arXiv:2602.04519, 2026.

Xizhou Zhu, Jian Wang, Yifan Chen, et al. Evomaster: A foundational agent framework for building evolving autonomous scientific agents at scale. arXiv preprint arXiv:2601.08912, 2026.

A Evidence-Admission Hyperparameters and Sensitivity Analysis

Table 4 summarizes the default hyperparameter values for the Triage admission gate. All five parameters are fixed across all benchmark tasks and backbones without task-specific tuning.

Table 4: Evidence-admission gate hyperparameters. All values are fixed across all experiments.

Constant	Role	Value
τ_{anom}	Residual threshold to open diagnostic state	0.85
ϵ_{rep}	Same-channel agreement tolerance	0.12
ϵ_{ind}	Cross-channel agreement tolerance	0.12
τ_{gen}	Held-out gain required to authorize a revision	0.16
k	Independent held-out records required for acceptance	2

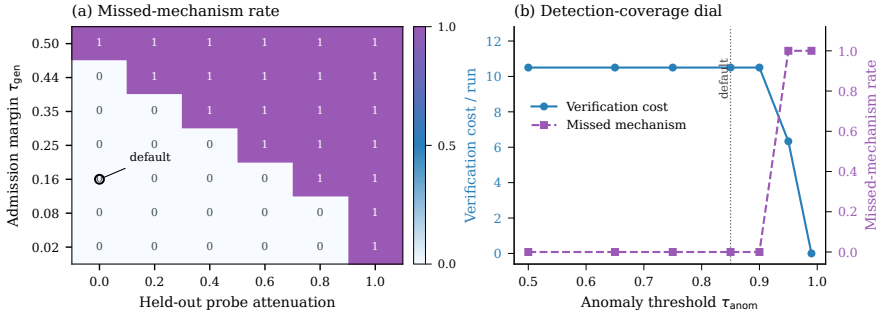


Figure 6: Gate sensitivity analysis. (a) Missed-mechanism rate as a joint function of the held-out gain threshold τ_{gen} and held-out probe attenuation. The default operating point (circle) lies in the zero-error region, and the boundary moves monotonically with effective held-out gain. (b) Verification cost and missed-mechanism rate as a function of the anomaly threshold τ_{anom}; the default (dashed) lies on the flat zero-miss plateau before coverage collapses at more restrictive thresholds.

Figure 6 presents the two-dimensional sensitivity phase map across decision thresholds. The default held-out threshold ($\tau_{\text{gen}} = 0.16$) lies safely within the zero-error region, requiring over $\sim$80% signal attenuation before a true mechanism is mistakenly rejected.

B Exploration Diversity (APD) and Exploitation Trajectories (AUOC)

Candidate exploration diversity (Average Pairwise Distance, APD) and cumulative exploitation throughput (Area Under the Optimization Curve, AUOC) across all three backbone LLMs are reported in Table 2 in the main text. Figure 4 plots the task-specific best-so-far convergence trajectories under Qwen3-235B.

C Execution Latency and Wall-Clock Throughput Profiles

Figure 8 details wall-clock execution latency and scientific discovery throughput across all eight baselines under Qwen3-235B. Triage-PiEvo requires only 134.5s per run (+13.6% overhead over standard PiEvo), avoiding the heavy latencies of multi-agent debate or tree search while achieving the highest exploitation throughput (71.2% AUOC).

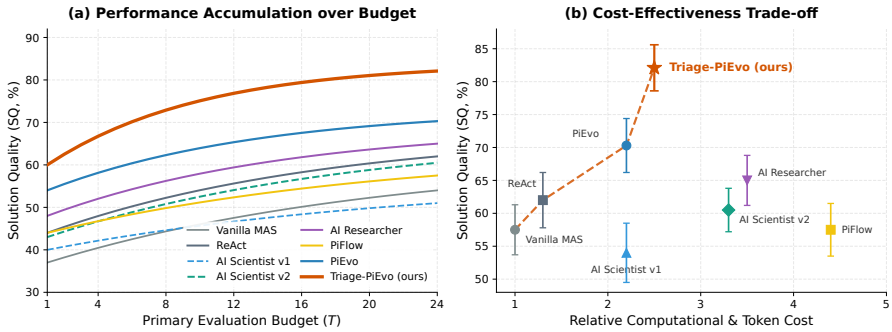


Figure 7: Performance progression and cost-effectiveness trade-off. (a) Solution Quality (SQ, %) over budget $T \in [1, 24]$. (b) Cost-effectiveness trade-off: Solution Quality versus relative compute and token cost, defining the empirical efficiency frontier.

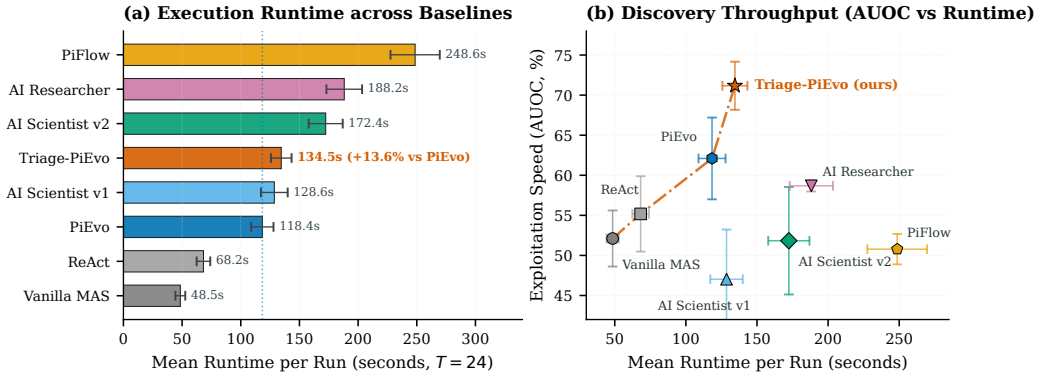


Figure 8: Execution runtime and discovery throughput across baselines. (a) Mean wall-clock runtime per 24-step discovery run across all eight baselines under Qwen3-235B. Triage-PiEvo incurs only +13.6% overhead over standard PiEvo due to conditional anomaly-triggered verification, while remaining substantially faster than tree-search or multi-agent debate frameworks (AI Scientist v2, AI Researcher, PiFlow). (b) Scientific exploitation throughput: Exploitation Speed (AUOC, %) versus mean runtime (seconds). Triage-PiEvo achieves the peak discovery throughput, reaching the highest AUOC (71.2%) at competitive execution latency.

D Cross-Harness Benchmarks and Triage as a Plug-in Module

We evaluate on the optimization tasks from AutoResearchEval (Fei et al., 2026), a 100-task benchmark spanning seven domains and the research lifecycle. Table 5 benchmarks Triage-PiEvo against four autonomous agent harnesses: Claude Code, Codex CLI, AI-Researcher (Tang et al., 2025), and ResearchHarness. Among the compared systems, Triage-PiEvo obtains the highest Official Score (0.418 ± 0.028) and Target Attainment (0.386). While the comparison agents show a gap between raw official score and externally verified score, Triage-PiEvo attains a Verified Score of 0.382. It also reduces Critical Failures to 5.2% (compared with 14.8% for Claude Code and 26.2% for ResearchHarness), with 86.5% valid runs and a verification cost of $0.92 per task.

Triage as a Plug-in Module. To test whether the benefits of evidence admission are tied specifically to the PiEvo candidate generator or transfer across host architectures, Table 6 evaluates Triage as a verification layer on top of three harnesses (ResearchHarness, AI-Researcher, and Claude Code), comparing Base execution, standard Self-Review prompting, and our + Triage gate. Within this evaluation, adding Triage yields absolute Verified Score

Table 5: Performance on AutoResearchEval Optimization Tasks. Comparison of Triage-PiEvo against autonomous scientific and coding agent harnesses across benchmark tasks. Official Score entries show empirical mean ± SD over independent seeds; the other columns report means. Bold indicates the best result.

Method	Official Score ↑	Target Attainment ↑	Verified Score ↑	Valid Runs (%) ↑	Critical Failure (%) ↓	Cost / Task ($) ↓
Claude Code	0.348 ± 0.032	0.312	0.298	78.5%	14.8%	0.84
Codex CLI	0.315 ± 0.041	0.274	0.261	69.2%	22.4%	1.12
AI-Researcher	0.332 ± 0.036	0.295	0.284	74.1%	18.6%	0.68
ResearchHarness	0.286 ± 0.045	0.241	0.230	65.8%	26.2%	0.75
Triage-PiEvo (Ours)	0.418 ± 0.028	0.386	0.382	86.5%	5.2%	0.92

gains of 0.098 on ResearchHarness, 0.087 on AI-Researcher, and 0.097 on Claude Code. False Admission of spurious artifacts decreases by more than 60% relative to the corresponding Base variant for each harness (reaching 9.6% on Claude Code), while Revision Accuracy reaches 78.2%, with additional token and runtime cost. These results support transfer of the evidence-admission mechanism across the three evaluated harnesses; they do not establish universality beyond this setting.

Table 6: Triage as a Plug-in across Autonomous Research Harnesses. Incorporating Triage enhances task score and verified reliability while reducing false evidence admissions across the three host architectures.

Harness	Variant	Task Score ↑	Verified Score ↑	False Admission (%) ↓	Revision Accuracy (%) ↑	Token Cost ↓	Runtime (s) ↓
ResearchHarness	Base	0.286	0.230	38.4%	44.2%	14.2k	48.5
	+ Self-Review	0.301	0.248	35.2%	48.5%	18.6k	62.1
	+ Triage (Ours)	0.345	0.328	14.2%	68.4%	21.4k	71.3
AI-Researcher	Base	0.332	0.284	32.6%	51.8%	16.5k	56.2
	+ Self-Review	0.348	0.305	29.8%	56.2%	22.1k	74.8
	+ Triage (Ours)	0.389	0.371	11.8%	73.6%	25.8k	86.5
Claude Code	Base	0.348	0.298	29.4%	55.4%	18.8k	64.0
	+ Self-Review	0.365	0.321	26.8%	60.2%	25.4k	85.2
	+ Triage (Ours)	0.412	0.395	9.6%	78.2%	29.2k	98.4

E Systematic Ablation of Evidence Admission Mechanisms

Table 7 evaluates each component of the Triage admission policy: the confirmation threshold $k \in \{1, 2, 3\}$, held-out verification, channel reliability tracking, recursive backtracking on retraction, and treating a same-source repeat as independent. Two comparisons carry the argument. Removing held-out verification raises false revision from 4.8% to 34.5%, and treating a same-source repeat as independent raises it to 45.8%—both admit evidence that is reproducible but not transferable, which is the artifact case. Removing channel reliability raises false revision to 19.2% through a different route: verification actions are then routed without regard to which channels have been reconcilable, so corroboration is sought from channels already in disagreement. The gate therefore depends on both the standard of proof and the choice of who supplies it.

Table 7: Ablation of Evidence Admission Mechanisms. Systematic evaluation of each architectural component in Triage . The full framework achieves the optimal balance between high mechanism recall, high verification accuracy, and low false revision risk.

Variant	False Revision (%) ↓	True Mechanism Recall (%) ↑	Verified Score ↑	Retraction Acc. (%) ↑	Verification Cost ($) ↓
Full Triage (Ours)	4.8%	92.4%	0.418	93.5%	0.24
$k = 1$ (Permissive Gate)	22.4%	89.2%	0.362	78.5%	0.14
$k = 3$ (Strict Gate)	6.5%	71.8%	0.370	90.8%	0.38
w/o Held-out Verification	34.5%	83.5%	0.315	61.2%	0.18
w/o Channel Reliability	19.2%	81.2%	0.352	76.3%	0.22
w/o Backtracking	16.5%	84.0%	0.338	51.8%	0.21
Same-Source Repeat = Independent	45.8%	78.0%	0.265	42.1%	0.20

F Algorithmic Specification of the TRIAGE Decision Protocol

Algorithm 1 formalizes the executable decision loop of the Triage evidence-admission gate.

Algorithm 1 The Triage Evidence-Admission Protocol

Require: Candidate x_t, observed primary outcome y_t, primary prediction $\hat{y}_t$, primary channel c, active principles $\mathcal{P}_t$, threshold τ_{anom}, agreement tolerances $\epsilon_{\text{rep}}, \epsilon_{\text{ind}}$, held-out margin τ_{gen}, confirmation threshold $k = 2$.

Compute anomaly residual: $s_t \leftarrow |y_t - \hat{y}_t|/\sigma_t$.
if $s_t \leq \tau_{\text{anom}}$ then
  Admit (x_t, y_t) to primary search history $\mathcal{H}_{\text{search}}$; return $\mathcal{P}_t$.
end if
{Stage 1 — same-channel repeat. Controls admission to search only.}
$v_t^c \leftarrow \text{Evaluate}(x_t, \text{repeat}, c)$ {same channel c as the primary observation}
if $|y_t - v_t^c| \leq \epsilon_{\text{rep}}$ then
  $\alpha_c \leftarrow \alpha_c + 1$; admit (x_t, y_t) to $\mathcal{H}_{\text{search}}$ as search-active.
else
  $\beta_c \leftarrow \beta_c + 1$; substitute the verified median $y_t \leftarrow v_t^c$ and admit to $\mathcal{H}_{\text{search}}$.
  return $\mathcal{P}_t$ {unreconciled observation: no revision path}
end if
{A same-channel repeat never authorizes revision, however often it agrees:}
{a reproducible artifact agrees with itself. Revision requires Stage 2.}
if a principle revision proposal P is generated from (x_t, y_t) then
  {Stage 2a — cross-channel check against a distinct channel}
  Select verifier channel $c' \leftarrow \arg\max_{c'' \neq c} p_{c''}^{\text{trust}}$ among declared independent channels.
  $v_t^{c'} \leftarrow \text{Evaluate}(x_t, \text{cross-channel}, c')$
  if $|v_t^c - v_t^{c'}| > \epsilon_{\text{ind}}$ then
    $\beta_c \leftarrow \beta_c + 1$; return $\mathcal{P}_t$ {channels disagree: source unresolved}
  end if
  $\alpha_c \leftarrow \alpha_c + 1$; set state of P to Provisional.
  {Stage 2b — held-out generalization, the only admitting test}
  Draw $X_{\text{heldout}} \sim \mathcal{D}_{\text{heldout}}$ with $X_{\text{heldout}} \cap \mathcal{H}_{\text{search}} = \emptyset$, fixed before the run.
  $E(P) \leftarrow \{e = (x', y') : x' \in X_{\text{heldout}},\ e.\text{ind} = \text{true}\}$, with each stance set by the τ_{gen} margin of §3.
  if $\exists\, e \in E(P)$ with $\text{contradict}(e, P) = 1$ then
    Set state of P to Retracted; recursively retract every principle listing P as parent.
  else if $|\{e \in E(P) : e.\text{ind} \wedge e.\text{heldout} \wedge \text{support}(e, P)\}| \geq k$ then
    Set state of P to Accepted; $\mathcal{P}_{t+1} \leftarrow \mathcal{P}_t \cup \{P\}$.
  else
    Leave P Provisional {support below k: search continues, state unchanged}
  end if
end if
return $\mathcal{P}_{t+1}$.

G Extended Benchmark Protocols and Failure Metric Definitions

AutoResearchEval Benchmark. AutoResearchEval (Fei et al., 2026) evaluates multi-step targeted optimization across molecular properties, material bandgaps, and algorithmic code generation. Each harness is run across 5 independent seeds with an identical per-task monetary cap ($2.00) and time budget.

Failure Metric Definitions.

- Official Score: Raw target optimization metric reported by the host framework upon run termination.
- Target Attainment: Binary indicator indicating whether the best discovered candidate achieves $\geq 90\%$ of the task-specific global target optimum.
- Verified Score: Score evaluated independently by an external, sandboxed ground-truth testbed.

- Valid Runs (%): Percentage of runs that complete all execution rounds without unhandled runtime crashes, tool timeouts, or syntax failure.
- Critical Failure (%): Percentage of runs in which unverified, hallucinated tool outputs or corrupted feedback poisoned the agent's internal memory, leading to permanent loop collapse or degenerate repeating actions.